

+24p

Convergence of Newton-Cotes Quadrature

+12p

J. L. Clements – mdlout project, Sydney – james.l.clements.iii@gmail.com

Abstract

We revisit the composite trapezoidal and Simpson 1/3 rules and chart their behaviour on four representative integrals, two smooth and two with endpoint singularities. On smooth integrands we recover the textbook $O(h^2)$ and $O(h^4)$ rates; on singular integrands both rules degrade together, and the high-order advantage of Simpson's rule collapses to a small constant factor. The poster summarises the results in three columns: introduction, method, and numerical evidence.

Introduction

Numerical integration is the inner loop of most scientific computing. Closed-form antiderivatives are the exception; the rule is to discretise the integration interval and sum weighted samples of the integrand. The two sim-

in modern guises whenever a new class of integrand puts pressure on existing libraries.

Method

Let $f : [a, b] \rightarrow \mathbb{R}$ and partition $[a, b]$ into n equal subintervals of width $h = (b - a)/n$. The composite trapezoidal rule approximates the integral by

$$T_n(f) = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

with leading error $-(b-a)h^2 f''(\xi)/12$ for $\xi \in (a, b)$, provided $f \in C^2[a, b]$. The rule is second-order accurate.

With n even, Simpson's 1/3 rule groups nodes in overlapping triples and integrates the quadratic interpolant exactly:

$$S_n(f) = \frac{h}{3} \left[f(x_0) + 4 \sum_{k=1}^{n/2} f(x_{2k-1}) + 2 \sum_{k=1}^{n/2-1} f(x_{2k}) + f(x_n) \right]$$

with leading error $-(b-a)h^4 f^{(4)}(\eta)/180$ for $\eta \in (a, b)$. The rule is fourth-order accurate. Both rules were implemented in double precision IEEE 754 with Kahan-compensated summation [3]

n	index
right	
8	index
right	
16	index
dent right	
32	index
dent right	
64	index
dent right	

The convergence of absolute error is -2.00 and $\ln$ prediction

[SVG file omitted]

Log-log of Simpson's

Discussion

Three textbook observations and one subtler one fall out. First, on smooth integrands Simpson's rule should be the default: the implementation cost is essentially identical to the trapezoidal rule and the asymptotic advantage is dramatic. Second, the asymptotic theory is fragile – once smoothness is lost at an endpoint, both rules degrade together. Third, adaptive routines interleave the two rules not for accuracy but for *error estimation*, exploiting Richardson extrapolation on the Simpson-minus-trapezoidal difference.

The subtler observation is that the Simpson-to-trapezoidal error ratio tends to a function of the integrand's regularity, not of h . Students who memorise $h^4 \ll h^2$ are often surprised when, on a real singular integrand, the two rules agree to within a factor of three.

ture and asymptotic expansions. *Mathematics of Computation*, 21(98):162-178, 1967.

References

References

1. R. Piessens et al. *QUADPACK: A Subroutine Package for Automatic Integration*. Springer-Ver-